

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I Divyasree.M/192524098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Divyasree. M*

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
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|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated | | | |

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

Problem 1 — Retail Data Optimization

1.1 Industry Context

A retail chain running both physical stores and an e-commerce platform generates continuous, high-volume, heterogeneous data: point-of-sale transactions, web/app clickstreams, RFID/shelf-sensor feeds, and CRM/loyalty history. The business goals are threefold — (a) real-time inventory visibility across stores, warehouses and the online channel to prevent stock-outs and overstock; (b) demand forecasting to drive replenishment and supply-chain planning; and (c) personalized marketing that increases conversion and customer lifetime value. The key architectural challenge is unifying fast-moving operational data (needed within seconds for inventory) with large historical datasets (needed for forecasting and personalization models trained over months of behaviour).

1.2 Proposed Architecture

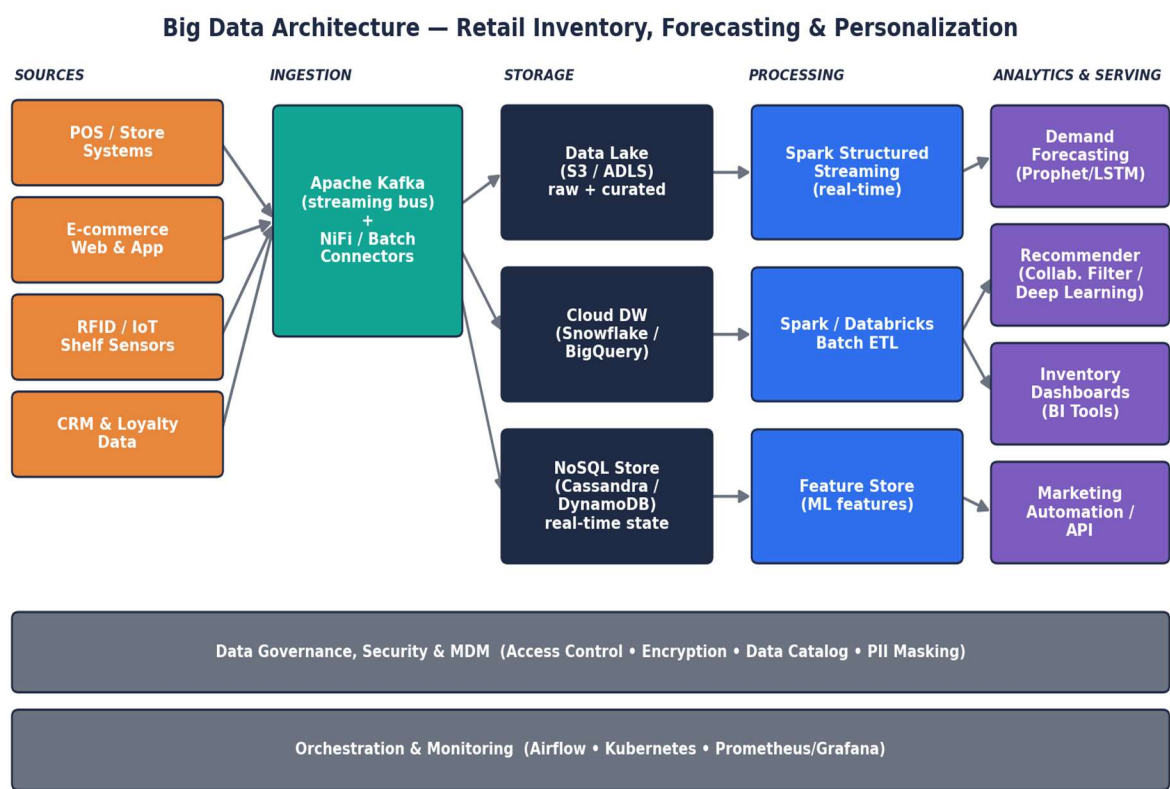


Figure 1. Lambda-style architecture combining a real-time inventory path with a batch analytics path feeding forecasting and recommendation engines.

The design follows a Lambda-style pattern: a speed layer streams POS, e-commerce and sensor events through Kafka into Spark Structured Streaming, updating a low-latency NoSQL store (Cassandra/DynamoDB) that powers real-time inventory dashboards and stock-out alerts. In parallel, a batch layer lands the same raw events into a cloud data lake (S3/ADLS), from which scheduled Spark/Databricks jobs curate data into a cloud data warehouse (Snowflake/BigQuery) for reporting, forecasting and recommendation-model training. A shared feature store keeps the two paths consistent for machine learning.

1.3 Tools & Technologies

Layer / Stage	Technology / Tool	Purpose
Ingestion	Apache Kafka, Apache NiFi	Capture streaming POS/e-commerce events and batch feeds from ERP/CRM systems
Storage	Amazon S3 / Azure Data Lake, Snowflake or BigQuery, Cassandra/DynamoDB	Raw/curated data lake, analytical warehouse, and low-latency operational store for live stock counts
Processing	Apache Spark (batch), Spark Structured Streaming	ETL/curation, real-time aggregation of sales and inventory movements
Orchestration	Apache Airflow, Kubernetes	Schedule ETL/ML pipelines and run services at scale
BI & Visualization	Power BI / Tableau / Looker	Store manager and merchandising dashboards

1.4 Analytics & Machine Learning Models

- Demand forecasting: time-series models (Prophet, ARIMA/SARIMA) for stable-demand SKUs and LSTM/temporal deep-learning models for products with promotions or seasonality.
- Inventory optimization: reorder-point and safety-stock calculations driven by forecast error distributions; anomaly detection on shrinkage/stock discrepancies.
- Personalization: collaborative filtering and content-based recommenders for product suggestions; customer segmentation (RFM/clustering) to target campaigns; uplift modelling to identify customers most responsive to promotions.
- Real-time triggers: streaming rules that flag low-stock SKUs and automatically notify replenishment systems or store staff.

1.5 Security, Privacy & Operational Risk

- PII protection: tokenize/mask customer identifiers used in loyalty and recommendation pipelines; apply role-based access control (RBAC) across lake, warehouse and BI layers.
- Regulatory compliance: align personalization and marketing data use with data-protection regulation (e.g., GDPR/DPDP-style consent and right-to-erasure handling).
- Operational resilience: schema validation at ingestion to prevent bad data from corrupting inventory counts; monitoring/alerting (Prometheus/Grafana) on pipeline latency, since stale inventory data directly causes stock-outs or overselling.
- Data quality governance: a central data catalog and master data management (MDM) for product/SKU identifiers to keep store and online inventory reconciled.

Problem 2 — Social Media Fraud & Misinformation Detection

2.1 Industry Context

A large social media platform processes billions of daily interactions — posts, comments, likes, shares and new account signups. Malicious actors create fake/bot accounts and coordinate to spread misinformation, so detection must operate in near real time to limit the spread and reach of harmful content before it goes viral, while minimizing false positives that would wrongly restrict genuine users. The core challenges are extreme event volume/velocity, the need for graph-aware reasoning (fake accounts act in networks, not isolation), and combining automated action with human review for contested cases.

2.2 Proposed Architecture

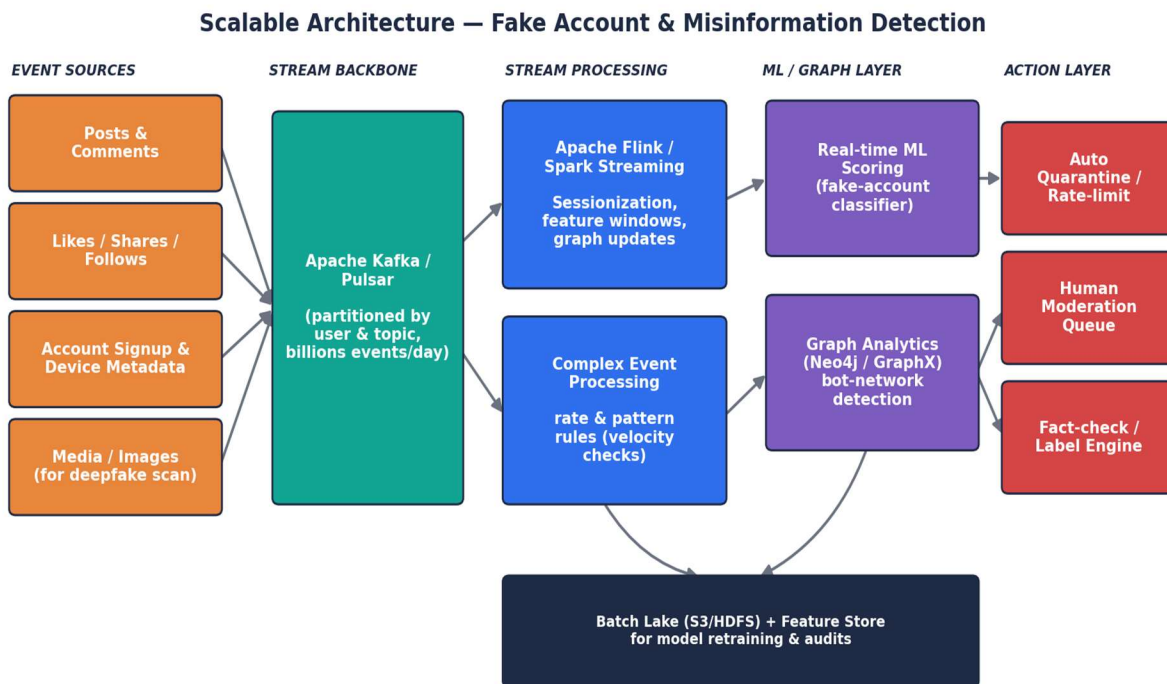


Figure 2. Streaming pipeline with ML scoring and graph analytics feeding an automated + human-in-the-loop action layer.

All user interactions and account-signal events flow into a high-throughput distributed log (Kafka or Pulsar), partitioned by user/topic to sustain billions of events per day. Apache Flink performs stateful stream processing — sessionization, rolling behavioural features and rule-based velocity checks (e.g., accounts posting at superhuman rates). A real-time ML scoring layer classifies individual accounts/content, while a parallel graph-analytics layer (Neo4j or Spark GraphX) detects coordinated bot networks and inauthentic amplification by analysing follow/share/like graphs. High-confidence detections are auto-quarantined or rate-limited; ambiguous cases are routed to a human moderation queue; content flagged as likely misinformation is passed to a fact-checking/labelling engine. All events also land in a batch lake for model retraining and audit.

2.3 Tools & Technologies

Layer / Stage	Technology / Tool	Purpose
Streaming backbone	Apache Kafka / Apache Pulsar	Ingest and distribute billions of interaction events per day with partitioned, ordered delivery
Stream processing	Apache Flink, Spark Streaming	Low-latency feature computation, session windows, rule-based anomaly checks
Graph analytics	Neo4j, Spark GraphX / GraphFrames	Detect bot networks and coordinated inauthentic behaviour via graph connectivity
ML serving	TensorFlow Serving / Triton, feature store	Host and serve real-time fake-account and content-risk classifiers
Storage	S3/HDFS data lake	Long-term storage for retraining, audits and regulatory reporting

2.4 Role of Machine Learning & Stream Processing

- Stream processing frameworks (Flink/Spark Streaming) provide the low-latency, stateful computation needed to compute rolling behavioural features (posting velocity, engagement ratios) within seconds of an event occurring — essential because fraud/misinformation spreads exponentially if undetected early.
- Supervised ML classifiers (gradient-boosted trees or deep networks) score accounts/content using engineered features (account age, network centrality, content similarity to known misinformation, device fingerprint reuse).
- Graph neural networks / graph algorithms (community detection, PageRank-style centrality) identify clusters of coordinated fake accounts that individual-account scoring would miss.
- NLP models (transformer-based classifiers, embedding similarity search) detect near-duplicate misinformation narratives spreading across many accounts.

2.5 Security, Privacy & Operational Risk

- False-positive management: human-in-the-loop review for medium-confidence cases to avoid wrongly suspending genuine users; maintain an appeals workflow.
- Adversarial robustness: continuously retrain models since bad actors adapt; monitor for concept drift and coordinated evasion attempts.
- Data privacy: interaction graphs contain sensitive relationship data — restrict access to graph/ML systems, encrypt data in transit and at rest, and apply least-privilege access for moderation teams.
- Transparency & compliance: maintain audit trails of automated actions to support platform transparency reporting and regulatory obligations around content moderation.

Problem 3 — Government National Data Platform

3.1 Industry Context

A government agency is building a national data platform that integrates census, economic and geospatial datasets contributed by multiple departments, each with its own systems, formats and ownership. Departments need shared, interoperable access to support policy analysis, while the platform must guarantee strict privacy and security for citizen-level data. Unlike the other four problems, the central challenge here is less about raw processing speed and more about governance: agreeing shared data standards, controlling access across organizational boundaries, and ensuring long-term data quality and accountability.

3.2 Proposed Governance & Architecture

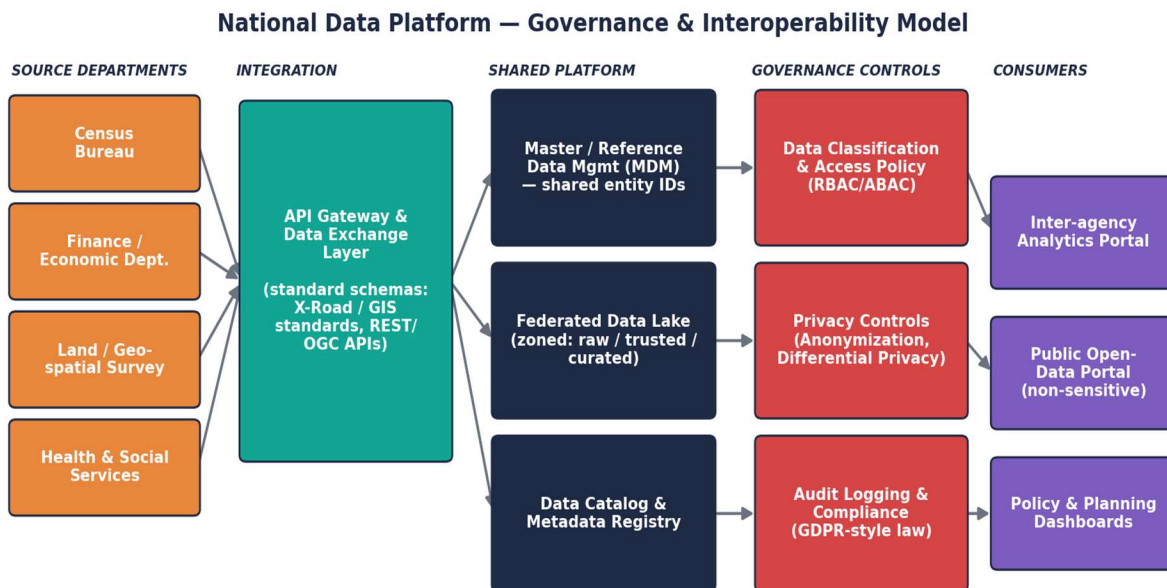


Figure 3. Federated integration layer with centralized governance controls mediating access between source departments and consumers.

Each department retains its own source systems; a standardized API gateway/data-exchange layer (built on open interoperability standards such as REST/OGC for geospatial data and a national data-exchange protocol) mediates all sharing. Integrated data lands in a federated data lake organized into raw, trusted and curated zones, backed by master data management (MDM) so the same citizen, business or geographic entity is consistently identified across datasets. A data catalog and metadata registry make data assets discoverable. Access is governed centrally through classification policies, privacy-preserving techniques and full audit logging before any inter-agency analytics portal, public open-data portal, or policy dashboard can consume the data.

3.3 Governance & Data Management Strategy

- Data governance council: a cross-department body that defines shared schemas/standards (e.g., common geographic and demographic identifiers), approves new data-sharing agreements, and resolves data-ownership disputes.

- Interoperability standards: adopt open, machine-readable standards (OGC for geospatial, common statistical data/metadata exchange formats) so datasets from different departments can be joined without bespoke integration for every pair of systems.
- Master data management (MDM): maintain golden records and shared entity identifiers (e.g., a unique geography code) so census, economic and geospatial datasets can be linked accurately.
- Data classification & access tiers: classify datasets (public, restricted, confidential/PII) and enforce role- and attribute-based access control (RBAC/ABAC) so departments only see what they are authorized to see.
- Privacy-by-design: apply anonymization, aggregation and differential privacy techniques before publishing to the public open-data portal; keep identifiable data behind the restricted tier only.
- Data quality & stewardship: assign data stewards per department accountable for accuracy, timeliness and documentation; automated validation rules at ingestion.
- Auditability & compliance: comprehensive audit logging of every access/query for regulatory compliance and public accountability, with periodic third-party security audits.

3.4 Tools & Technologies

Layer / Stage	Technology / Tool	Purpose
Integration	API Gateway (Kong/Apigee), Enterprise Service Bus	Standardized, monitored data exchange between departments
Storage	Federated data lake (Hadoop/S3) with zoned architecture	Raw, trusted and curated zones supporting multiple data domains
Governance tooling	Data catalog (Collibra/Apache Atlas), MDM platform	Metadata management, lineage tracking, entity resolution
Privacy tooling	Differential privacy libraries, data masking/tokenization	Protect citizen-level data before wider sharing or publication
Access & audit	Identity & Access Management (IAM), SIEM logging	Enforce RBAC/ABAC and maintain a full audit trail

3.5 Security & Risk Awareness

- Strong authentication (multi-factor) and network segmentation between departments and the shared platform.
- Encryption at rest and in transit for all shared datasets; key management under a dedicated security team.
- Regular privacy impact assessments before onboarding new datasets, particularly where census/economic data could be re-identified when combined with geospatial data.

Problem 4 — Banking Fraud Detection

4.1 Industry Context

A bank processes millions of financial transactions every minute across cards, online/mobile banking, ATMs and wire transfers. Fraud must be detected and, where necessary, blocked within milliseconds to avoid financial loss, while minimizing false declines that inconvenience legitimate customers. The solution must combine long-term historical transaction patterns (to build accurate customer/merchant profiles) with sub-second streaming analysis (to catch fraud as it happens), making this one of the most latency-sensitive big data use cases.

4.2 Proposed Architecture

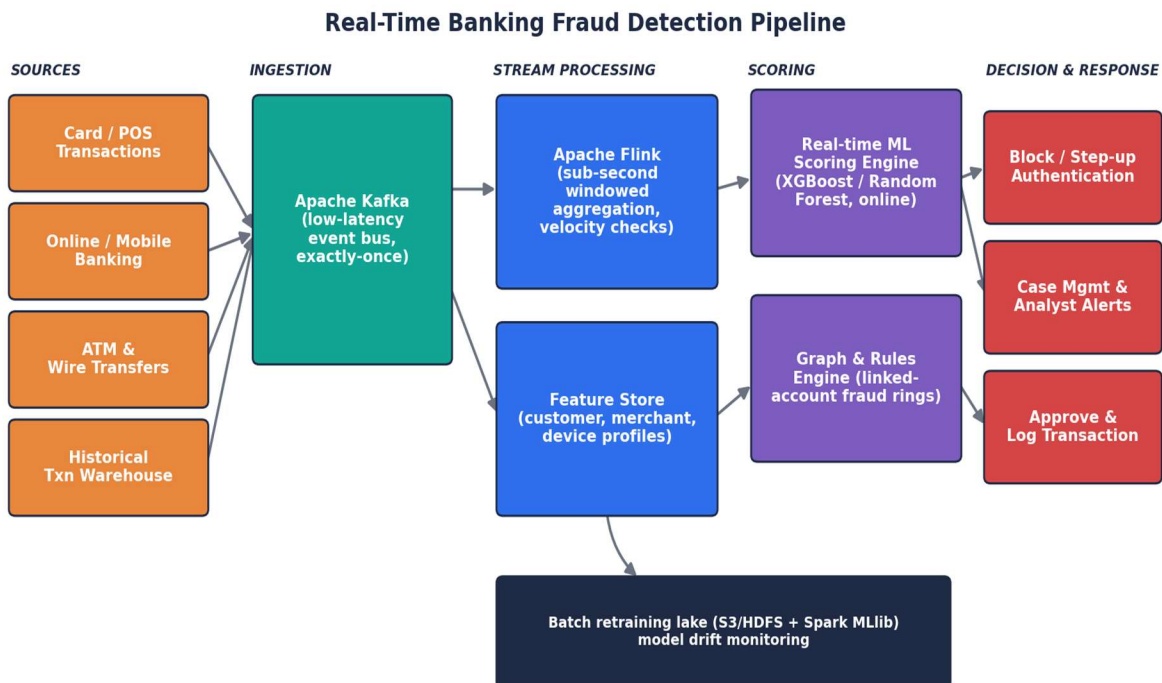


Figure 4. Low-latency streaming pipeline with real-time ML scoring, feature store, and a batch retraining loop.

Transactions from card/POS, online banking, ATM and wire-transfer systems are published to Kafka configured for low-latency, exactly-once delivery. Apache Flink performs sub-second windowed aggregation (e.g., transaction velocity, geographic impossibility checks) and joins each event against a feature store holding customer, merchant and device profiles built from historical data. A real-time ML scoring engine (e.g., gradient-boosted trees such as XGBoost, or online-learning models) assigns a fraud-risk score, while a parallel graph/rules engine detects fraud rings via linked accounts, devices or beneficiaries. Based on the combined score, the system blocks or requests step-up authentication, raises an analyst alert, or approves and logs the transaction. All transactions and outcomes feed a batch lake for periodic model retraining and drift monitoring.

4.3 Tools & Technologies

Layer / Stage	Technology / Tool	Purpose
Ingestion	Apache Kafka (low-latency, exactly-once config)	Reliable, ordered delivery of transaction events at bank scale
Stream processing	Apache Flink	Sub-second aggregation, windowed features, velocity/geo checks
Feature store	Feast or custom feature store	Consistent real-time and batch features for customer/merchant/device
ML scoring	XGBoost/Random Forest (online serving), deep learning for complex patterns	Real-time fraud probability scoring per transaction
Graph/rules engine	Neo4j / rules engine (Drools)	Detect fraud rings via shared devices, accounts or beneficiaries
Batch/retraining	Spark MLlib on S3/HDFS lake	Periodic retraining and model-drift monitoring

4.4 Analytics Models & Algorithms

- Supervised classification (XGBoost, Random Forest, or deep neural networks) trained on labelled historical fraud/non-fraud transactions for the primary risk score.
- Unsupervised anomaly detection (isolation forests, autoencoders) to catch novel fraud patterns not present in historical labels.
- Graph-based link analysis to identify fraud rings sharing devices, IP addresses or beneficiary accounts.
- Rule-based velocity/geo-impossibility checks as a fast, interpretable first line of defence layered beneath the ML score.
- Continuous online/incremental learning or scheduled retraining to adapt to evolving fraud tactics and reduce model drift.

4.5 Security, Latency & Risk Considerations

- Low-latency design: in-memory feature lookups and stream-native scoring keep the end-to-end decision under the ~100ms budget typically required at point of authorization.
- High accuracy vs. customer experience trade-off: calibrate thresholds to balance fraud loss against false-decline rate; use step-up authentication for medium-risk cases instead of outright blocking.
- Encryption & tokenization: card and account numbers are tokenized before entering the analytics pipeline; strict PCI-DSS-aligned controls on all systems touching payment data.
- Model governance: explainability (e.g., SHAP values) for regulatory and dispute-resolution purposes; regular model risk-management review as required by banking regulators.

Problem 5 — Smart Farming / Precision Agriculture

5.1 Industry Context

A smart farming system collects soil, weather and crop data from distributed sensors and drones across potentially remote fields with limited connectivity. Farmers want practical, timely insights to improve yield and optimize resource use (water, fertilizer, pesticide). The architectural challenge is handling data from many geographically distributed, intermittently connected edge devices and converting it into actionable, low-latency guidance that a non-technical farmer can act on.

5.2 Proposed Architecture

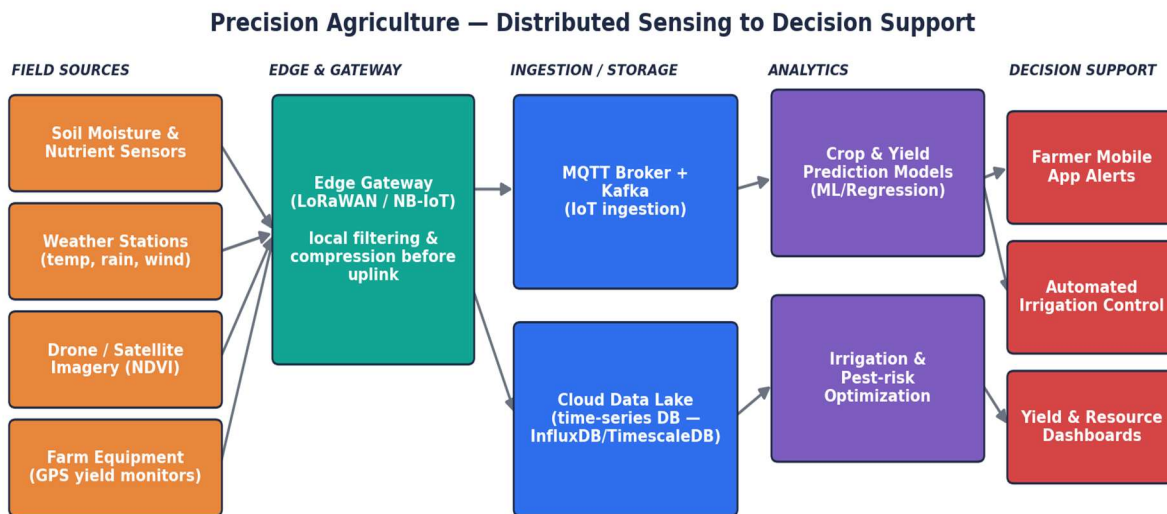


Figure 5. Edge-to-cloud IoT architecture for precision agriculture, from field sensors through analytics to farmer-facing decision support.

Soil, weather, drone/satellite imagery and equipment-GPS data are collected at the field edge and pre-processed by an edge gateway (using low-power wide-area protocols such as LoRaWAN or NB-IoT) to filter and compress data before uplink — important given limited rural bandwidth. Data is published via MQTT/Kafka into a cloud time-series database (InfluxDB/TimescaleDB) alongside a broader data lake for imagery and equipment logs. Analytics services generate crop/yield predictions and irrigation/pest-risk recommendations, which are delivered back to the farmer through a mobile app, can trigger automated irrigation control, and are summarized in yield/resource dashboards.

5.3 Tools & Technologies

Layer / Stage	Technology / Tool	Purpose
Edge/IoT	LoRaWAN/NB-IoT gateways, edge compute (Raspberry Pi/industrial IoT nodes)	Local sensor aggregation and pre-filtering before uplink, coping with limited connectivity
Ingestion	MQTT broker, Apache Kafka	Lightweight IoT messaging feeding a scalable event backbone
Storage	Time-series DB (InfluxDB/TimescaleDB), cloud data lake for imagery	Efficient storage/query of high-frequency sensor readings and drone/satellite imagery
Analytics/ML	Spark/Python ML stack, computer vision for NDVI imagery	Yield prediction, crop health/pest-risk detection, irrigation optimization
Delivery	Mobile app, IoT actuators for irrigation control	Deliver insights and, where appropriate, automate irrigation directly

5.4 Analytics Supporting Decision-Making

- Yield prediction: regression/ML models combining historical yield, soil nutrient levels and weather data to forecast expected output and guide input planning.
- Crop health monitoring: computer-vision analysis of drone/satellite NDVI imagery to detect stress, disease or nutrient deficiency before it is visible to the naked eye.
- Irrigation optimization: models combining soil-moisture sensor data with weather forecasts to recommend precise irrigation timing/volume, reducing water waste.
- Pest & disease risk scoring: combining weather patterns (humidity/temperature) with historical outbreak data to give early warnings.
- Resource dashboards: aggregate views letting farmers compare fields/seasons and track input costs against yield outcomes.

5.5 Security & Operational Risk

- Device security: authenticate and encrypt data from field sensors/gateways to prevent tampering with agronomic data that drives automated irrigation decisions.
- Connectivity resilience: edge devices buffer and retry uploads during network outages common in rural areas, avoiding data loss.
- Data ownership & privacy: farm-level data (yields, practices) is commercially sensitive — ensure farmers retain ownership/control and that data sharing with agritech vendors is consent-based.
- Safety of automation: human-override controls on any automated irrigation/actuation to prevent crop damage from a faulty sensor reading or model error.